

TOLEDO CGS, OH, USA

WMO: 997295

Lat: 41.694N

Lon: 83.473W

Elev: 604

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.8	(c) 7.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.5	(k) 30.4	(l) 23.4	(m) 26.2	(n) 8.0	(o) 260	(p) 0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.4	(c) 90.5	(d) N/A	(e) 87.5	(f) N/A	(g) 84.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.1	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.3	(b) 20.6	(c) 18.3	(e) -2.4	(f) 95.0	(g) 7.8	(h) 3.0	(i) -8.0	(j) 97.2	(k) -12.6	(l) 98.9	(m) -16.9	(n) 100.6	(o) -22.6	(p) 102.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	26.4	27.2	36.8	49.9	61.1	70.6	74.8	72.9	65.8	54.6	41.7	32.9		
	DBStd	19.14	11.10	10.08	9.88	8.67	8.60	6.67	5.83	5.19	7.21	8.00	9.04	8.92		
	HDD50	2766	734	641	429	105	7	0	0	0	0	40	277	533		
	HDD65	5909	1198	1060	874	459	183	22	2	4	75	337	700	995		
	CDD50	3258	1	1	20	101	352	619	769	709	473	182	27	4		
	CDD65	927	0	0	0	6	63	190	306	248	99	15	0	0		
	CDH74	6724	0	0	3	68	448	1339	2524	1655	603	84	0	0		
	CDH80	2025	0	0	0	6	121	429	826	439	186	18	0	0		
(14) Wind	WSAvg	8.7	9.9	9.3	9.2	10.7	9.0	8.6	7.3	6.9	7.0	8.1	9.1	9.3		
(15) Precipitation	PrecAvg	33.6	1.9	1.8	2.4	3.2	3.5	3.6	3.4	3.4	3.1	2.4	2.4	2.4		
	PrecMax	46.7	3.6	4.8	5.7	5.0	6.1	8.9	6.3	8.3	6.3	6.3	6.7	4.0		
	PrecMin	24.4	0.6	0.4	0.6	1.3	0.6	1.5	1.2	0.7	0.4	0.4	0.9	1.1		
	PrecStd	5.2	0.9	1.1	1.3	1.2	1.7	1.8	1.3	1.5	1.4	1.3	1.3	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	57.3	71.3	80.7	88.0	93.2	94.8	91.3	91.0	82.3	68.2	60.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	51.6	50.2	63.5	75.1	83.7	89.1	90.9	87.8	85.6	75.4	62.8	54.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	47.1	44.2	57.9	69.8	79.5	85.5	87.7	85.0	80.7	71.1	59.2	50.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	41.4	39.6	51.7	64.8	74.8	81.5	84.6	82.0	76.8	67.0	55.5	45.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		11.7	11.7	13.1	15.0	14.9	13.8	14.4	13.8	13.8	13.8	12.1	9.9		
	5% DB	MCDBR	16.6	17.4	21.0	23.4	20.1	18.7	17.8	17.2	19.0	18.6	18.0	15.5		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.335	0.360	0.405	0.435	0.451	0.444	0.435	0.387	0.362	0.324	0.314		
	taud		2.411	2.297	2.318	2.218	2.182	2.184	2.225	2.234	2.346	2.399	2.465	2.402		
	Ebn at Noon		269	274	280	274	268	263	264	262	268	262	259	252		
	Edh at Noon		25	33	36	43	46	46	44	42	35	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		487	773	1107	1468	1721	1920	1927	1688	1372	904	576	397		
	RadStd		50	102	101	124	148	129	106	62	111	74	77	38		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	N/A	

Nomenclature: See separate page